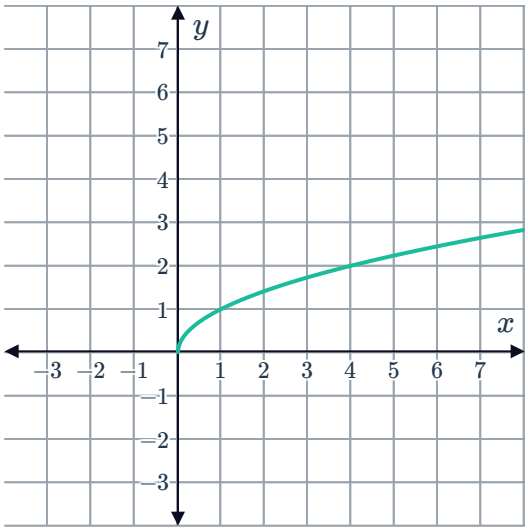
Graphs of square root functions

1

a No

b ∞

c



d As x increases, the function increases at a slower and slower rate.

e $(0, 0)$

f No

g No

h $y = 5\sqrt{x}$ increases more rapidly than $y = \sqrt{x}$.

2

a No

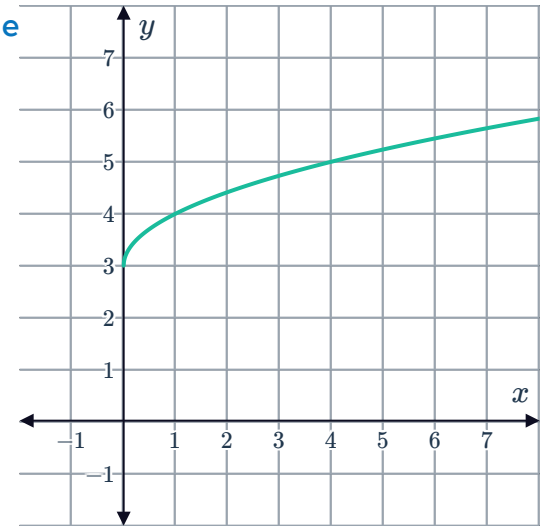
b The function decreases at a slower and slower rate.

3

a No **b** ∞ **c** $y = 3$ **d** 0

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e



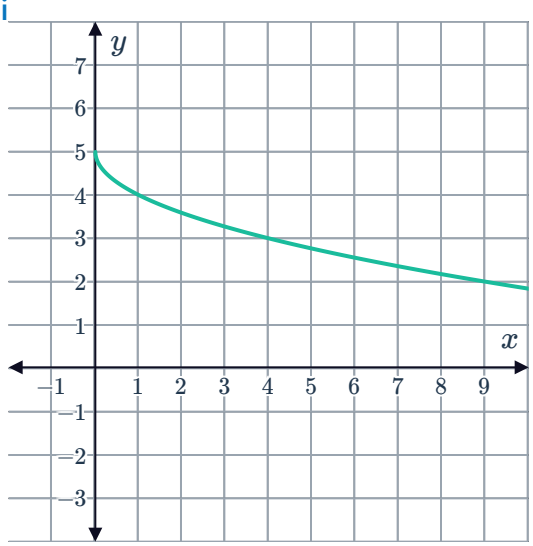
4

a $x \leq 0$ **b** $y \geq 0$ **c** ∞

5

a

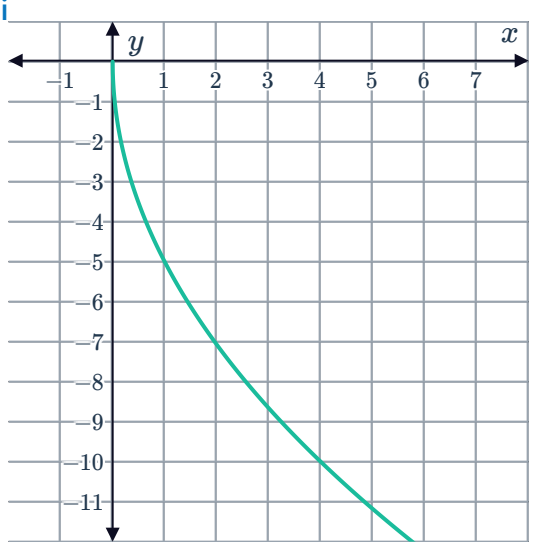
i $x \geq 0$ **ii** $y \leq 5$ **iii**



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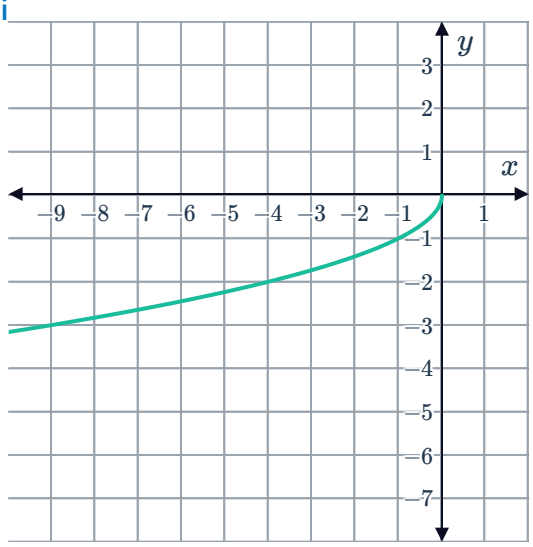
b

i $x \geq 0$ **ii** $y \leq 0$ **iii**



c

i $x \leq 0$ **ii** $y \leq 0$ **iii**



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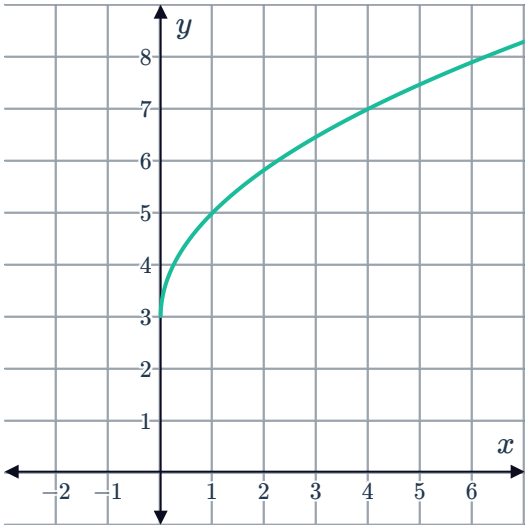
6

a Increasing

b More steep

c $(0, 3)$

d



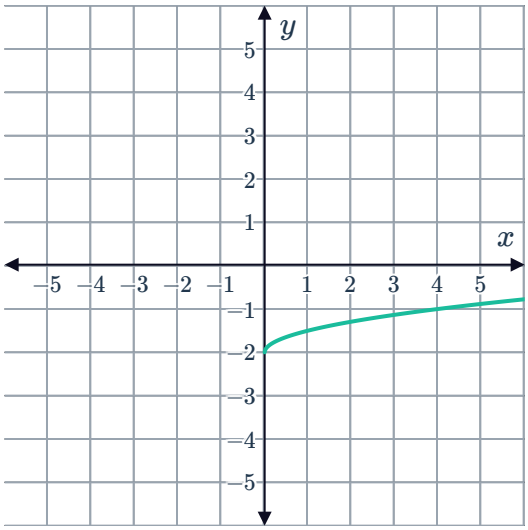
7

a Increasing

b Less steep

c $(0, -2)$

d



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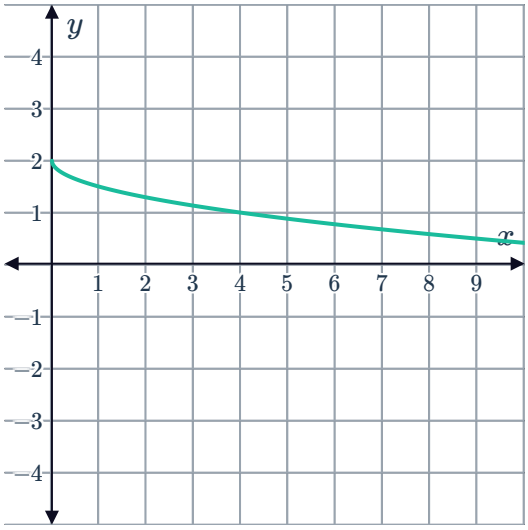
8

a Decreasing

b Less steep

c $(0, 2)$

d



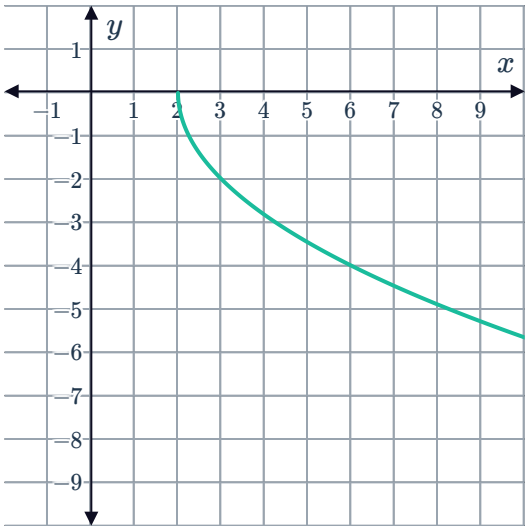
9

a Decreasing

b More steep

c $(2, 0)$

d



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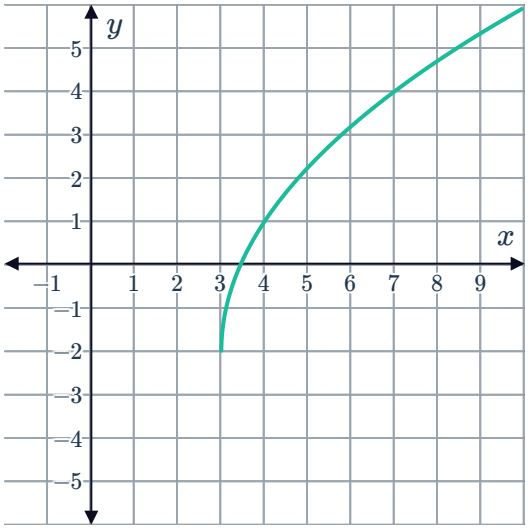
10

a Increasing

b More steep

c $(3, 2)$

d



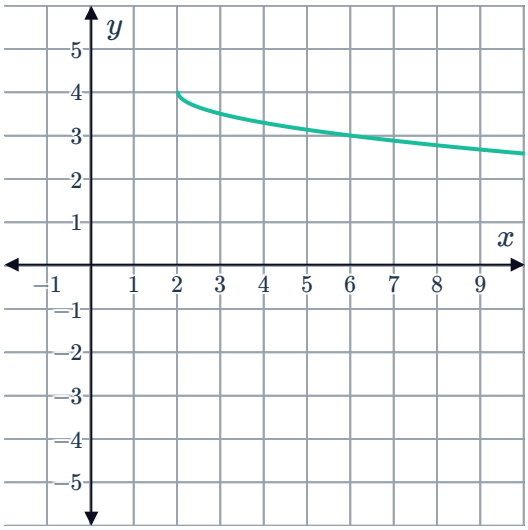
11

a Decreasing

b Less steep

c $(2, 4)$

d

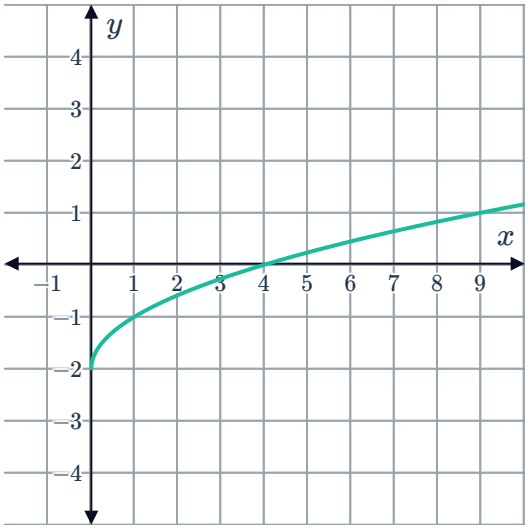


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12

a

i

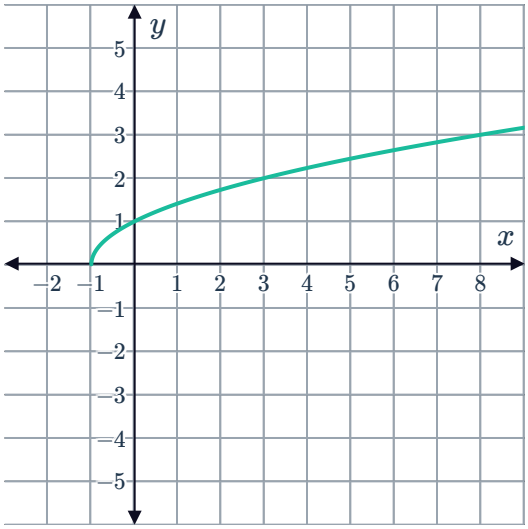


ii $x \geq 0$

iii $y \geq -2$

b

i

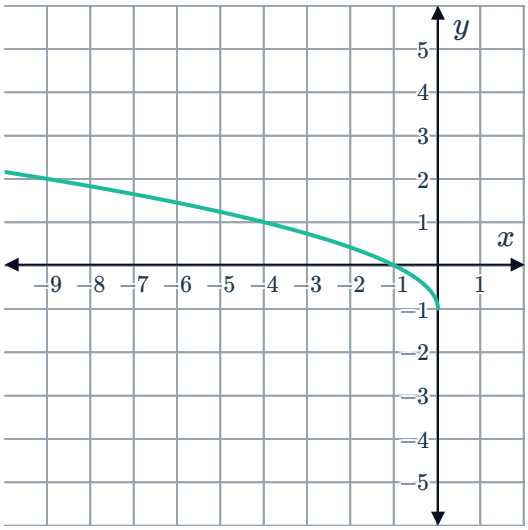


ii $x \geq -1$

iii $y \geq 0$

c

i

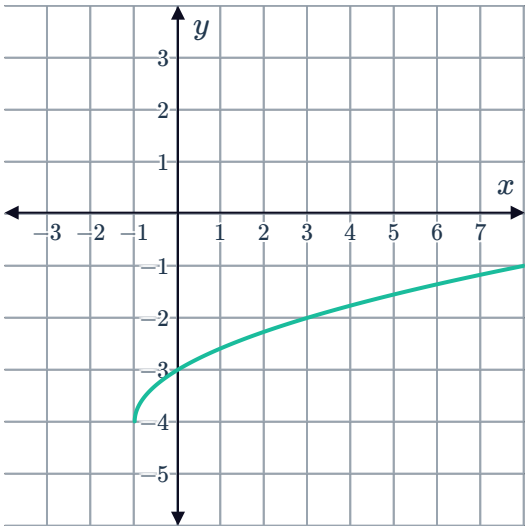


ii $x \leq 0$

iii $f(x) \geq -1$

d

i



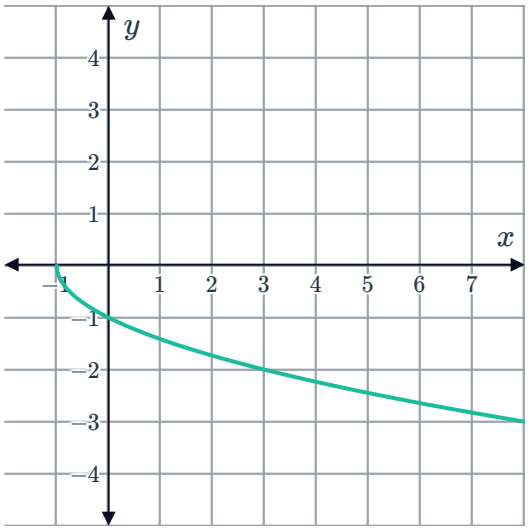
ii $x \geq -1$

iii $y \geq -4$

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e

i

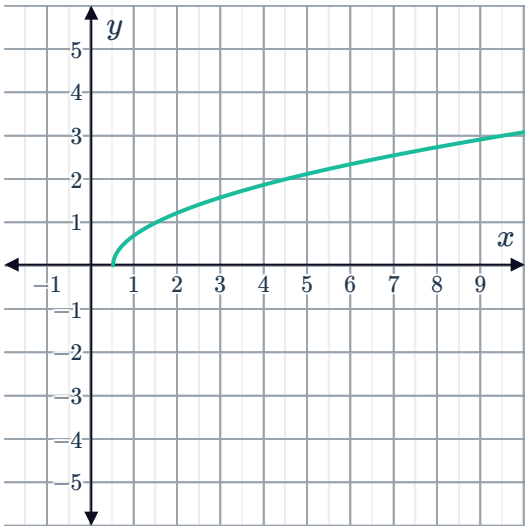


ii $x \geq -1$

iii $y \leq 0$

f

i

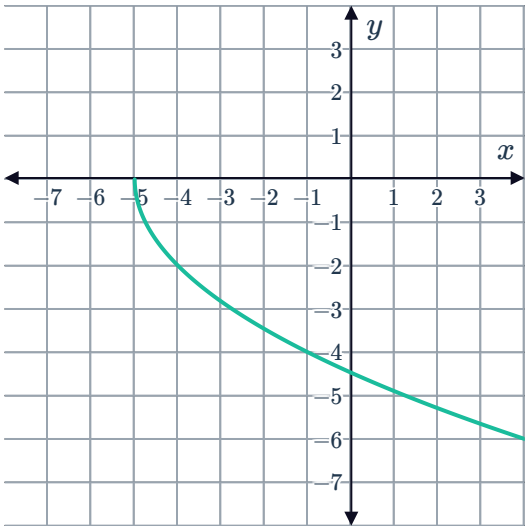


ii $x \geq 2$

iii $y \geq 0$

g

i

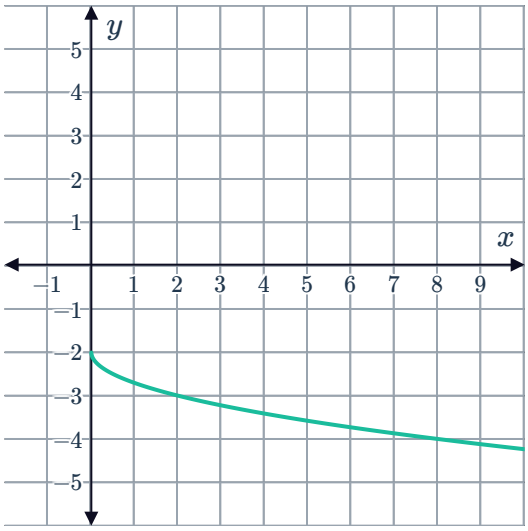


ii $x \geq -5$

iii $y \leq 0$

h

i



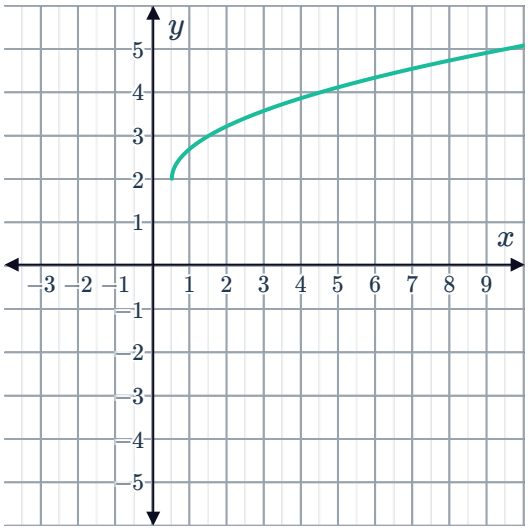
ii $x \geq 0$

iii $y \leq -2$

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i

i



ii $x \geq 1$

iii $y \geq 2$

Characteristics of square root functions

13

a No **b** No **c** No **d** No

e Yes

14 Domain: $x \geq 0$, range: $y \geq -2$

15

a $x \geq 5$ **b** $y \geq 0$

c

i Yes **ii** No **iii** No

16

a 6 **b** $x \leq 0$ **c** $y \geq 6$

17 $[2, \infty)$

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Transformations

18

a $y = 2$

b Vertical shift 2 units up

c $f(x) = \sqrt{x} + 2$

19

a $x = -2$

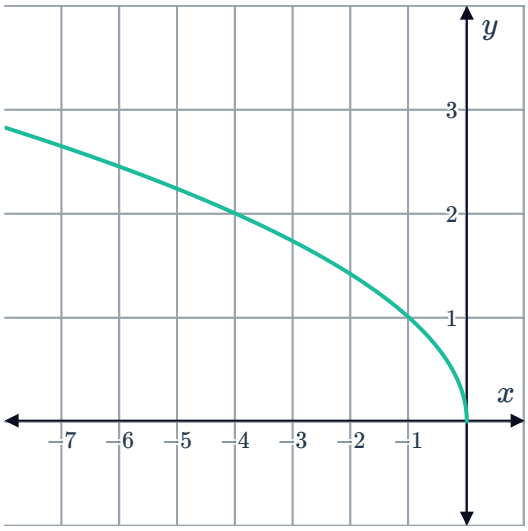
b Horizontal shift 2 units to the left

c $f(x) = \sqrt{x+2}$

20

a

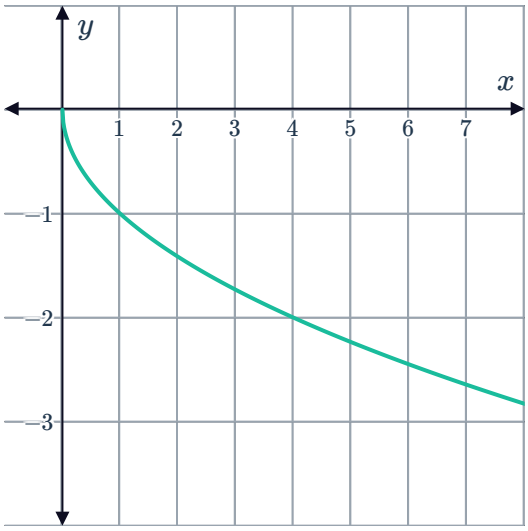
i



ii $y = \sqrt{-x}$

b

i



ii $y = -\sqrt{x}$

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21

a $x \leq 0$

b $y = 3\sqrt{-x}$

22

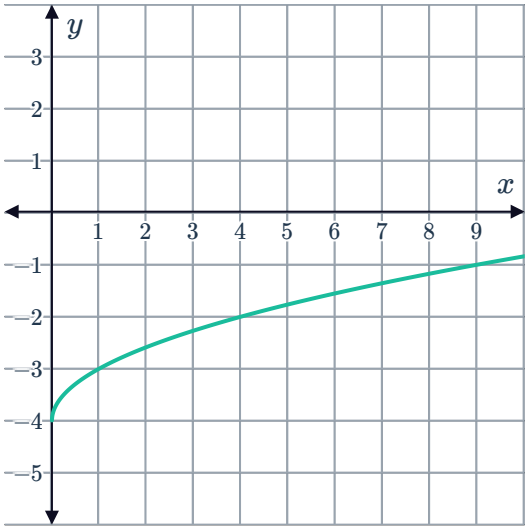
a 3

b Downwards

23

a The original function was translated down 4 units.

b



24 y increases more rapidly on $y = 3\sqrt{x}$ than on $y = \sqrt{x}$

25

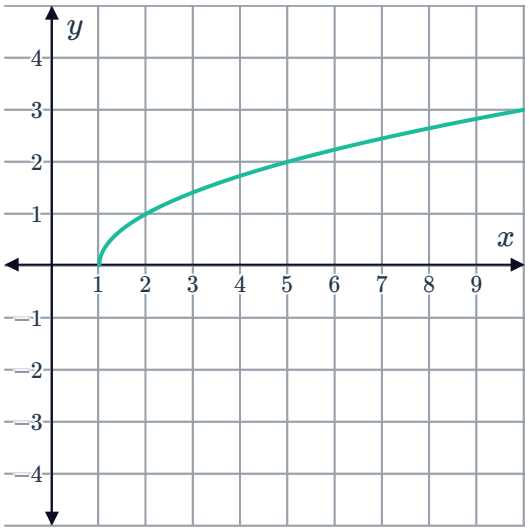
a $(0, -2)$

b $(0, 3)$

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26

a



b Horizontal translation 1 unit(s) to the right

27

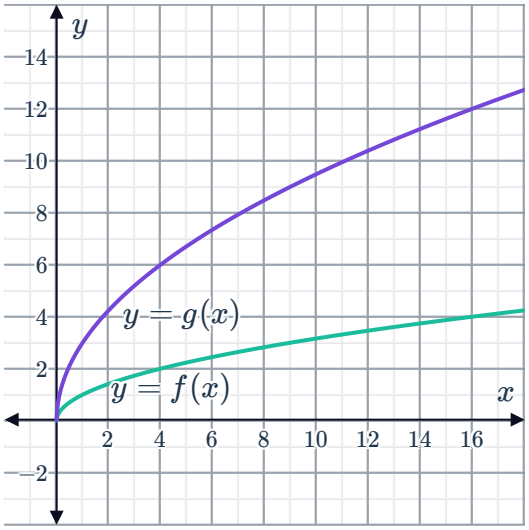
- Reflection across x-axis, then a translation 5 units down
- Translation 5 units up, then a reflection across x-axis

28

a

x	0	1	4	9	16
$y = f(x)$	0	1	2	3	4
$g(x) = 3f(x)$	0	3	6	9	12
$h(x) = -4f(x)$	0	-4	-8	-12	-16

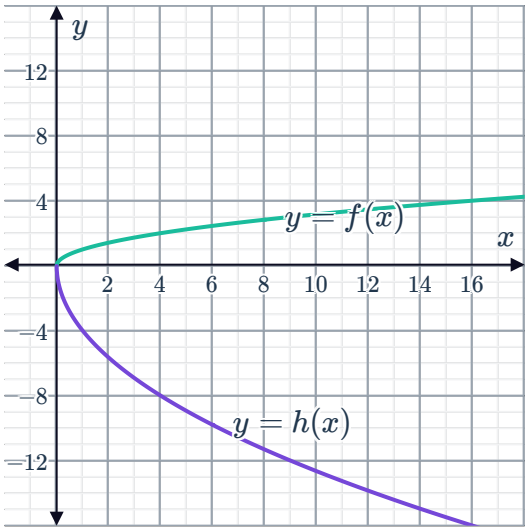
b



c Vertically stretched by a factor of 3

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d

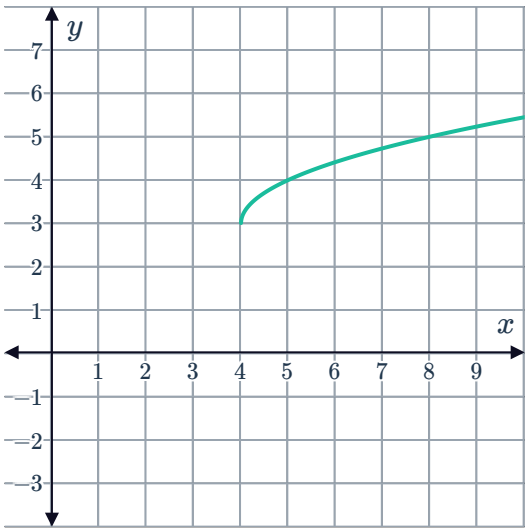


e Vertically stretched by a factor of 4 and reflected across the x-axis

29

a Translate the graph 4 units to the right and 3 units up.

b



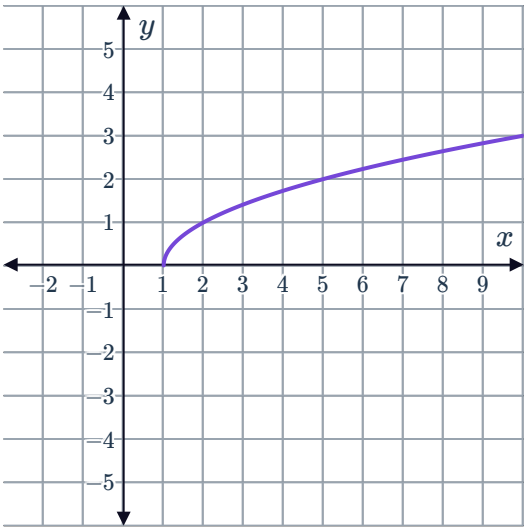
30

a $y = \sqrt{x - 5}$ **b** $y = -\sqrt{x} + 6$ **c** $y = -\frac{1}{2}\sqrt{x}$

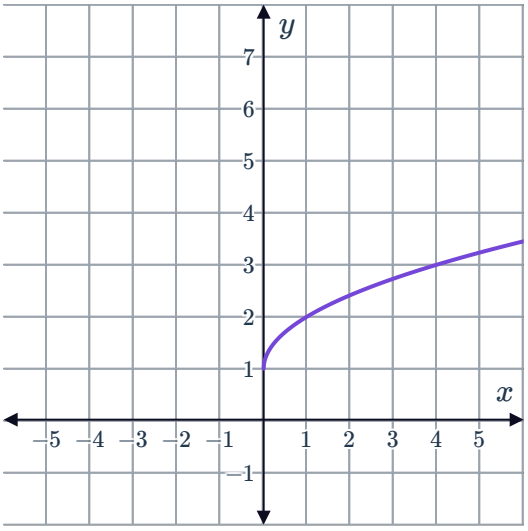
31 $\frac{\sqrt{2x}}{4} + 0$

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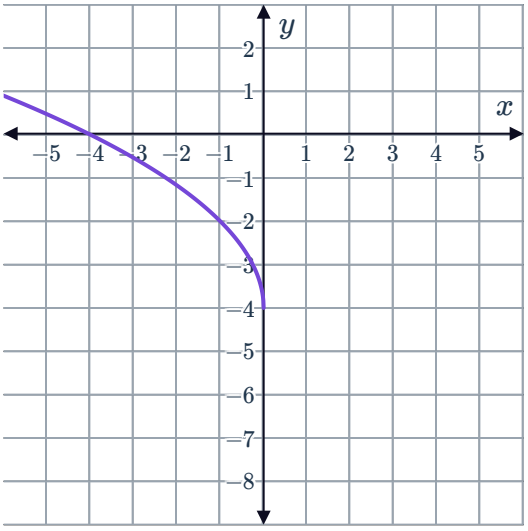
32



33



34



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Applications

35

a $4\sqrt{5}$ miles

b 8.9 miles

36 1.57 seconds

37

a $12\sqrt{2}$ feet per second

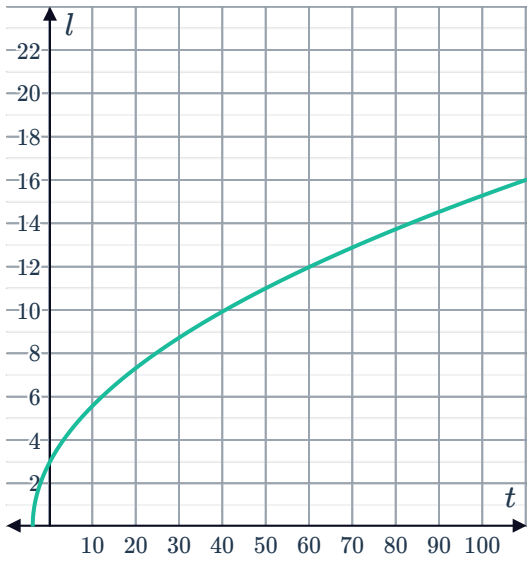
b 17 feet per second

38

a

Months (t)	0	60	96
Length (l)	3	12	15

b



c $t \geq 0$

d No

e 14.1 meters

f 0.7

39 $A = 14657415$

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40

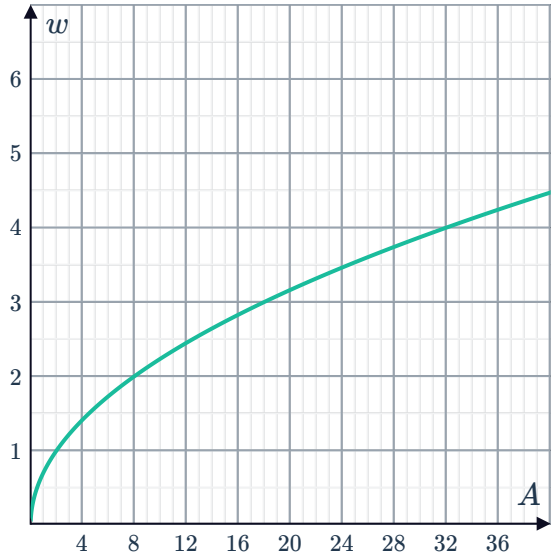
a $A = 2w^2$

b $w = \sqrt{\dfrac{A}{2}}$

c

Area (A)	0	2	8	18	32
Width (w)	0	1	2	3	4

d



e 50 units^2

f Above

41

a $D = -\sqrt{10}$ t

b $-5\sqrt{2}$ meters

c $\sqrt{2}$ m/s

d $2 - \sqrt{2}$ m/s

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